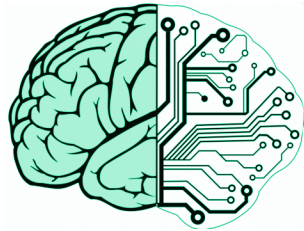


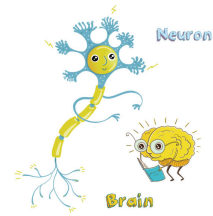
"Though this be madness, yet there is method in't."
Shakespeare: Hamlet: Act 2 Scene 2

The Hodgkin-Huxley neuron model V2.0



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Foreword

Hodgkin and Huxley's classic (V1.0) "model", which fundamentally changed our ideas about the physiology of neurons and which has served as the basis for describing the functioning of neurons for several decades, can hardly be overestimated; it is truly the crown jewel of neuroscience. However, its authors, quite realistically, assessed that the physical picture behind the observations they mathematically formulated could be partially incorrect. Already in their original communication, they drew attention to the fact that their equations did not fully describe their observations. Later they added that there exists observed evidence that does not fit into the picture. Due to the lack of certain knowledge, they were forced to formulate ad hoc assumptions and they admitted that the formalism of the equations could describe more than one physical process. Some of the assumptions proved to be a brilliant vision, but in quite a few cases it turned out that they did not always put the right physical background behind their observations which were generally correctly described mathematically, and that their model designed for mechanical computers had obvious limitations. Already at the time of the introduction of the model, doubts arose as to whether the purely electrical model could fully describe the observations known up to that time, and later, as the accuracy, directionality and detail of the observations increased, it became necessary to introduce more and more ad hoc assumptions in order to maintain the consistency of the model assuming purely electrical operation.

In the past decades, numerous (V1.0x) models have been created, modifying and supplementing this purely electrical, disciplinary, original model with more or less success (which ultimately became a model due to the activities of Hodgkin and Huxley (HH)'s followers). There have also been (V1.y) models that try to model the neuron within the framework of an entirely different discipline, mainly interpreting its thermodynamic, mechanical, optical, etc. phenomena. The common feature of all these models is that they cannot provide a complete model (describing all observations without contradiction). There are also models that do not even strive to describe a physical operating mechanism; they are satisfied with providing a mathematical formalism that shows behavior that is consistent with experience; without even trying to answer the questions 'why' and 'by what mechanism'.

Due to the lack of the needed expertise in physics, biophysics introduced the idea of the "whatnot" [1] protein mechanisms for the unexplainable forces that

totally misguided physiology and neuroscience. Biology has no laws of motion in the sense as physics has Newton's laws. The many speculative hypotheses hyde that science can describe living matter, but needs a different approach. Claims such as "pumps [by using protein mechanisms] . . . transport ions *against their electrical and chemical gradients*" [2], page 101, or "the flux of ions through ion channels is passive, requiring *no expenditure of metabolic energy* by the channels." [2], page 107, are as scientific as in the pre-Newtonian age, there was a notion related to Aristotelian philosophy. This concept gave credence to the "theory" that the celestial spheres and, later on, the planets were pushed or moved by celestial movers, celestial intelligences, or angels against physical forces, without continuously investing energy. Inventing the universal gravitational law made the complex interplay of those spheres obsolete, although, for the first look, it seemed to be unrelated.

Describing life science is still in the pre-Newtonian age: it has a lot of "spheres" described by complicated theories that contradict each other, and does not have its laws of motion, in the sense as Newton's Laws in physics. What is worse, it did not invent and test the laws describing living matter. Instead, it borrowed the disciplinary laws of physics, without recognizing that its different discipline has different conditions and requires different approximations; that is similar, but differently used laws. Without questioning the merits of the V1.0 model, the experimental experience accumulated in recent decades, the dizzying development of technology and instrumental measurements, and the ever-increasing demand for the results of neuroscience have made it necessary to develop the (V2.0) model. The model is necessarily cross-disciplinary; no single physical discipline alone allows for a complete and consistent interpretation of the phenomena of the living organism.

Chapter 1

Introduction

The *dynamic* operation of individual neurons, their connections, higher-level organizations, connections, the brain with its information processing capability, and finally, the mind with its conscience and behavior, are still among the big mysteries of science: *at which point the non-living matter becomes a living one, at which point the living matter becomes intelligent* and conscious; whether and how science can handle all this stuff. We really need a new understanding that we provide here. The hype in the newly launched projects is excessive and seems to lose the hope to build brain research on a firm science base. For example, the newly (at the end of 2025) launched “Brain/MINDS 2.0” program in Japan was launched using only mathematical models, *without targeting the understanding of the underlying physical processes*. In neuron models, ‘dynamics’ refers to how the state variables of the neuron (primarily membrane potential and ion channel properties) change and evolve over time in response to internal processes and external inputs. However, unlike in our approach, the usual processes simply use a time parameter in empirical or mathematical functions, see for an example [3]. We use a real dynamics, based on physics, in the sense as Newton introduced his laws of motion to describe response of neurons.

The book has several layers. In addition to the sections intended to be understandable by everyone (i.e. primarily readers with a suitable physiological and neuroscience background), there are also panels for those with other relevant backgrounds containing the necessary **mathematical 1.1** and **physical 1.1** knowledge, which require at least the appropriate level of BSc knowledge to understand, but sometimes knowledge that goes beyond this, and sometimes knowledge and explanations that make you delve deeper into the unusualness of the ideas. The complexity of the phenomenon requires many approximations, but by using the appropriate approximations and abstractions, the **numeric ??** panels contain numerical results (sometimes only reasonable estimates). We strive to make the material understandable without them, and with them, the reader with the appropriate qualifications will be able to gain knowledge that encourages further thought.

Physics panel 1.1

During the specialization of sciences, physics specialized in the description of inanimate nature and (in the interest of reasonable description) created fields of limited validity. In the study of living nature, its results, methods and tools were already used in a natural way. In doing so, they started from the correct point of view that the description of objects in nature and their interactions should be governed by the same laws. However, they did not take into account that due to the complexity of nature, approximations and neglects must be made in order to be able to describe them. Disciplinary laws create abstract concepts for a certain range of phenomena, and the laws describe these abstract concepts precisely, not the phenomena that occur in nature. In most cases these laws can be used in their entirety, due to the significant differences in the conditions of the living and non-living environment often they can only be used with significant modification or even complete rethinking; see the interpretation of Erwin Schrödinger's insightful questions in **Physics Panel 3.1**

Mathematics panel 1.1

This is mathematics

Numerics panel 1.1

This is numeric

Understanding the dynamic brain The *dynamic* operation of individual neurons, their connections, higher-level organizations, connections, the brain with its information processing capability, and finally, the mind with its conscience and behavior, are still among the big mysteries of science: *at which point the non-living matter becomes a living one, at which point the living matter becomes intelligent and conscious*; whether and how science can handle all this stuff. We really need a new understanding that we provide here. The

Quotation:

"The Human Brain Project should lay the technical foundation for a **new model** of ICT-based brain research, driving *integration between data and knowledge from different disciplines*, and catalysing a community effort to achieve a **new understanding of the brain**...and *new brain-like computing technologies*."

the Human Brain Project, summarised its goal @2012

hype in the newly launched projects is excessive and seems to lose the hope to build brain research on a firm science base. For example, the newly (at the end of 2025) launched "Brain/MINDS 2.0" program in Japan was launched using only mathematical models, *without targeting the understanding of the underlying physical processes*. In neuron models, 'dynamics' refers to how the state variables of the neuron (primarily membrane potential and ion channel properties) change and evolve over time in response to internal processes and external inputs. However, unlike in our approach, the usual processes simply use a time parameter in empirical or mathematical functions, see for an example [3]. We use a real dynamics, based on physics, in the sense as Newton introduced his laws of motion to describe response of neurons.

Abstract discussion Nature uses an infinite variety of implementing neurons. In the Central Nervous System (CNS) they can cooperate with each other, 'despite the extraordinary diversity and complexity of neuronal morphology and synaptic connectivity, *the nervous systems adopt a number of basic principles* for all neurons and synapses', [4] independently from the infinitely complex molecular, biochemical and physiological details. *We will base our holistic discussion on those general basic principles and create an 'abstract physical neuron'; a hypothetical element, which functions essentially like a neuron*, skipping the 'implementation details' nature uses. We essentially follow von Neumann's method in describing the foundations of computing science [5]: "we will base our considerations on a hypothetical element, which functions essentially like a neuron". We need different abstractions and approximations (furthermore, 'non-ordinary' laws of science) for describing biological processes. However, an abstraction is usable in practice only when paired with a generalization: the more abstract the assumption, the more general and widely applicable the concept or conclusion. *By abstractions, we can reduce the unbelievably detailed world into manageable pieces, and we can learn anything general*. We must show which approximations are oversimplifications and which phenomena are misunderstood; measured, or interpreted in a wrong approach.

Zigzag reading For the same reasons, we follow von Neumann's method when he described principles of technical computing [5]. "The ideal procedure would

be, to take up the specific parts in some definite order, to treat each one of them exhaustively, and go on to the next one only after the predecessor is completely disposed of. However, this seems hardly feasible. The desirable features of the various parts, and the decisions based on them, emerge only after a somewhat zigzagging discussion. It is, therefore, necessary to take up one part first, pass after an incomplete discussion to a second part, return after an equally incomplete discussion of the latter with the combined results to the first part, extend the discussion of the first part without yet concluding it, then possibly go on to a third part, etc. Furthermore, these discussions of specific parts will be mixed with discussions of general principles, of the elements to be used, etc.” For example, basic concepts such as the membrane potential and its regulation in resting and transients states, belong to chapter Physics, and their detailed scientific, physics-based, discussion is given there, but their corresponding aspects must be discussed in chapters ‘Abstract neuron’ and less abstract ‘Physiology’ as well. The contents are inherently intertwined, and the form must follow the contents. We make this zigzag reading more accessible by using hyperlinks and cross-references throughout the document.

Chapter 2

Modeling

Quotation:

”What I cannot create, I do not understand” R. P. Feynman

Quotation:

”Biological laws are much more difficult to find than physical ones” [6]
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2.1 Modeling methods

”There is a widely accepted distinction between merely modeling a mechanism’s behavior and explaining it. . . . Some models are data summaries. Some models sketch explanations but leave crucial details unspecified or hidden behind filler terms. Some models are used to conjecture a how-possibly explanation without regard to whether it is a how-actually explanation.” [7].

2.1.1 Disciplinary modeling

As Feynman R. P [8] wrote, ”the separation of fields [of science] . . . is merely a human convenience, and an unnatural thing. Nature is not interested in our separations, and *many of the interesting phenomena bridge the gaps between fields*”. ”When we go to investigate *we should not pre-decide what it is we are*

trying to do except to find out more about it. . . . The first principle is that you must not fool yourself, and you are the easiest person to fool.” Physiology, despite HH’s and Feynman’s warning, pre-decided that only the science discipline ‘electricity’ is authorized to describe neurons’ operation. However, electricity has no place for the mass and speed of its charge carriers. After seven decades, it should be realized, admitted, and fixed.

Quotation:

”Transient electrical signals are particularly important for carrying *time-sensitive information* rapidly and over long distances. These transient electrical signals—receptor potentials, synaptic potentials, and action potentials—are all produced by *temporary changes* in the electric current into and out of the cell, changes that drive the electrical potential across the cell membrane away from its resting value.” [4]

Chapter 3

Physics

3.1 Schrödinger on life

Quotation:

"Is life based on the laws of physics?" ... " *New laws to be expected in the organism*" ... E. Schrödinger: What is life?[1] @1944

Physics panel 3.1

"From all we have learnt about the structure of living matter, we must be prepared to find it working in a manner that cannot be reduced to the **ordinary laws of physics**. And *that not on the ground that there is any 'new force' or what not*, directing the behavior of the single atoms within a living organism, *but because the construction is different from anything we have yet tested in the physical laboratory.*" ... "How can the events in *space and time* which take place *within the spatial boundary* of a *living organism* be *accounted for by physics* and chemistry?" ... "living matter, while not eluding the 'laws of physics' as established up to date, is likely to involve 'other laws of physics' hitherto unknown, which, however, once they have been revealed, will form just as integral a part of this science as the former." (for the interpretation of those points, see **Physics panel3.2**) E. Schrödinger: What is life?[1] @1944

3.2 Disciplinarity

Physics panel 3.2

In his very accurately formulated question (see **Physics panel 3.1**), Schrödinger focused on (at least) these significant points

- '*events*' Unlike non-living matter, *living matter is dynamic, changing autonomously by its internal laws*; we must think differently about it, including making hypotheses and testing them in the labs. Those laws are non-ordinary 'because the construction is different', but its principles must not differ from the ones we already know.
- '*space and time*' Those characteristic points are significant changes resulting from processes that have material carriers, which change their positions with finite speed, so (unlike in classical mechanics) the events also have the characteristics 'time' in addition to their 'position'. In biology, the spatiotemporal behavior is implemented by *slow ion currents*. In other words, instead of 'moments' we must consider 'periods' and define the 'events' that define the beginning and the end of those periods.
- '*within the spatial boundary*' Laws of physics are usually derived for stand-alone systems, in the sense that the considered system is infinitely far from the rest of the world; also, in the sense that the changes we observe do not significantly change the external world, so its idealized disturbing effect will not change it. The 'construction' also results in constraint forces. Furthermore, we must consider the issue of finite resources.
- '*accounted for by physics*'[by extraordinary laws] We are accustomed to abstracting and testing a static attribute, and we derive the 'ordinary' laws of motion for the 'net' interactions. In the case of physiology, nature prevents us from testing 'net' interactions. We must understand that some interactions are non-separable, and we must derive 'extraordinary' laws. The forces are not unknown, but the known 'ordinary' laws of motion of physics are about single-speed interactions.
- '*living matter*' To describe its dynamic behavior, we must introduce a dynamic description. Processes happen inside it, and we can observe some characteristic points. The internal processes also manifest in changing physical quantities which are inseparable from the ones the experimenter intended to change.
- '*yet tested in the physical laboratory*'[including physiological ones] We test those 'constructions' in laboratories, in their actual environment, and in 'working state'. The measurement represents an

intervention into their internal processes that falsifies the measurement results. As we did with inanimate matter, we must develop and gradually refine the testing methods as well as the hypotheses. Not only in measuring them (including measurement design and evaluation) but also in handling them computationally, we need more or less different thinking and algorithms.

Version history

This is the documentation of the development of the site about neuronal operation, including the description of the unified theory of neuronal operation, its aspects in physics, physiology, computing and information theory and more. It discusses the simulator, which performs a time-aware simulation using the SystemC engine.

Versioning principles

The versions are ordered by the date of publishing. Only the middle versions are listed, the minor versions are only mentioned. Details are in the logs of the repository.

The contents are under development, and maintaining the versioning is of medium priority, so delays and

V0.2.x

Forces exerting on ions on the surface of the membrane added.

The version description was added. Text from the paper in "Foundations of Science" added.

V0.1.x

The consequences of science disciplinarity added

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